RACE
Regular Analysis Through Continuous Exercise

Thermodynamics 2 : RACE-01

Entropy and Spontaneity; Physical Significance of Entropy

1. **Assertion (A) :** A liquid crystallises into a solid and is accompanied by decrease in entropy.
Reason (R) : In crystals, molecules organise in an ordered manner.
(A) Both A and R are true and R is the correct explanation of A.
(B) Both A and R are true but R is not the correct explanation of A.
(C) A is true but R is false.
(D) A is false but R is true.

CTS166

2. The spontaneity means, having the potential to proceed without the assistance of external agency. The processes which occur spontaneously are
(A) flow of heat from colder to warmer body. (B) gas in a container contracting into one corner.
(C) gas expanding to fill the available volume. (D) burning carbon in oxygen to give carbon dioxide.

CTS200

3. For which of the following process ΔS°_{sys} is positive?
(A) The dissolution of ammonium nitrate in water.
(B) The expansion of a sample of chlorine into a larger volume.
(C) The decomposition of mercuric oxide.
(D) The rusting of iron.

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CTS186

Second Law of Thermodynamics

4. A system undergoes a process in which the entropy change is +5.51 JK^{-1}. During the process, 1.50 kJ of heat is added to the system at 300 K. The correct information regarding the process is
(A) the process thermodynamically reversible.
(B) the process is thermodynamically irreversible.
(C) the process may or may not be thermodynamically reversible.
(D) the process must be isobaric.

CTS187

5. List equation/law (in Column I) with statement (in Column II) :

Column-I		Column-II	
(A)	Arrhenius equation	(p)	Variation of enthalpy of a reaction with temperature
(B)	Kirchhoff equation	(q)	Variation of rate constant with temperature
(C)	Second law of thermodynamics	(r)	Entropy of an isolated system tends to increase and reach a maximum value
(D)	Hess's law of constant heat summation	(s)	Enthalpy change in a reaction is always constant and independent of the manner in which the

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6. Which of the following statement(s) is/are incorrect?

(A) Entropy is a measure of unavailable energy.

(B) Decrease in free energy of the system represents the unavailable energy of the system.

(C) Complete conversion of heat into work is not possible in a reversible isothermal expansion of ideal gas.

(D) Complete conversion of heat into work is not possible in an irreversible isothermal expansion of ideal gas.

CTS188

Entropy Change for an Ideal gas; Isothermal; Isochoric; Isobaric; Adiabatic Process

7. When one mole of an ideal gas is compressed to half of its initial volume and simultaneously heated to twice its temperature, the change in entropy is

(A) $C_{V,m}$ ln2 (B) $C_{P,m}$ ln2 (C) R ln2 (D) $(C_{V,m} - R)$ ln2

CTS189

8. The standard molar entropy of an ideal gas $\left(\gamma = \frac{4}{3}\right)$ is 2.5 cal/K mol at 25°C and 1 bar. The standard molar entropy of the gas at 323°C and 1 bar is (ln 2 = 0.7)

(A) undefined (B) 4R ln2 (C) 5.6 cal/K-mol (D) 8.1 cal/K-mol

CTS190

9. Molar heat capacity of CD_2O (deuterated form of formaldehyde) vapour at constant pressure is 14 cal/K-mol. The entropy change associated with the cooling of 3.2 g of CD_2O vapour from 1000 K to 900 K at constant pressure is (assume ideal gas behaviour for CD_2O) [ln 0.9 = −0.1]

(A) +0.14 cal/K (B) – 0.14 cal/K (C) −1.4 cal/K (D) +1.4 cal/K

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Entropy Change for Phase Transformation

10. The entropy change can be calculated by using the expression $\Delta S = \frac{q_{rev}}{T}$. When water freezes in a glass beaker, choose the correct statement amongst the following :

(A) ΔS (system) decreases but ΔS (surroundings) remains the same.

(B) ΔS (system) increases but ΔS (surroundings) decreases.

(C) ΔS (system) decreases but ΔS (surroundings) increases.

(D) ΔS (system) decreases and ΔS (surroundings) also decreases.

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ANSWER KEY

Question	1	2	3	4	5	6	7
Answer	A	CD	ABC	B	A → q, B → p, C → r, D → s	BCD	D
Question	8	9	10				
Answer	D	B	C				

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Thermodynamics 2 : RACE-02

Entropy Change for Phase Transformation

1. What is the entropy change when 3.6 g of liquid water is completely converted into vapours at 373 K? The molar heat of vaporization is 40.85 kJ/mol.
(A) 219.0 J/K (B) 2.190 J/K (C) 21.90 J/K (D) 109.5 J/K

CTS192

2. One mole of ice is converted into water at 273 K and 1 atm. The entropies of H_2O (s) and H_2O (l) are 38.0 and 58.0 J/K-mol, respectively. The enthalpy change for the conversion is
(A) 5460 kJ/mol (B) 5460 J/mol (C) −5460 J/mol (D) 20 J/mol

CTS193

Entropy Change for Chemical Reaction

3. Given the following entropy values (in J/K-mol) at 298 K and 1 atm $H_2(g)$ = 130.6, $Cl_2(g)$ = 223.0 and HCl(g) = 186.7. The entropy change (in J/K-mol) for the reaction $H_2(g) + Cl_2(g) \rightarrow 2HCl(g)$ is
(A) +540.3 (B) +727.0 (C) −166.9 (D) +19.8

CTS194

Gibb's Free Energy and Spontaneity; Calculation of Change in G (ΔG)

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4. Which of the following are correct at 298K
(A) $\Delta G°_F (H_{2(g)}) = 0$ (B) $\Delta_F G° (H_2O, l) = 0$ (C) $\Delta_F G° (D_2, g) = 0$ (D) $\Delta_F G° (H^+, aq.) = 0$

CTS168

5. Match the following parameters with description for spontaneity :

Δ (Parameters)				Description
	$\Delta_r H°$	$\Delta_r S°$	$\Delta_r G°$	
(A)	+	−	+	(i) Non-spontaneous at high temperature.
(B)	−	−	+ at high T	(ii) Spontaneous at all temperatures
(C)	−	+	−	(iii) Non-spontaneous at all temperatures

CTS169

6. The ΔG in the process of melting of ice at −15°C and 1 atm, is
(A) less than zero. (B) greater than zero. (C) equal to zero. (D) None of these

CTS195

7. The values of ΔG are very important in metallurgy. The ΔG values for the following reactions at 1000 K are given as follows.
$S_2(s) + 2O_2(g) \rightarrow 2SO_2(g)$; ΔG = −544 kJ
$2Zn(s) + S_2(s) \rightarrow 2ZnS(s)$; ΔG = −293 kJ
$2Zn(s) + O_2(g) \rightarrow 2ZnO(s)$; ΔG = −480 kJ
The ΔG for the reaction $2ZnS(s) + 3O_2(g) \rightarrow 2ZnO(s) + 2SO_2(g)$ will be
(A) −357 kJ (B) −731 kJ (C) −773 kJ (D) −229 kJ

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8. How much energy (in kJ) is available for sustaining muscular and nervous activity from the combustion of 1.0 mole of glucose molecules under standard conditions at 37°C (body temperature)? The standard entropy and enthalpy of reaction are +200 J/K-mol and – 2808 kJ/ mol, respectively.

CTS197

Carnot Cycle

9. An ideal gas undergoes through following cyclic process.

1–2 : Reversible adiabatic compression from $P_1V_1T_1$ to $P_2V_2T_2$

2–3 : Reversible isochoric heating from $P_2V_2T_2$ to $P_3V_3T_3$

3–4 : Reversible adiabatic expansion from $P_3V_3T_3$ to $P_4V_4T_4$

4–1 : Reversible isochoric cooling from $P_4V_4T_4$ to $P_1V_1T_1$

Efficiency of the cycles is :-

(A) $1-\left(\frac{T_4-T_1}{T_3-T_2}\right)$ (B) $1-\frac{T_1}{T_2}$ (C) $1-\left(\frac{V_2}{V_1}\right)^{\gamma-1}$ (D) $1-\frac{|Q_{4-1}|}{|Q_{2-3}|}$

CTS167

10. In which case will the efficiency of a Carnot cycle be higher, when the hot body temperature is increased by ΔT or when the cold body temperature is decreased by the same magnitude?

(A) Same in both.

(B) On decreasing the temperature of cold body.

(C) On increasing the temperature of hot body.

(D) Depends on the initial temperature of cold and hot bodies.

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11. What is the efficiency of a cycle consisting of two isochoric and two adiabatic lines, if the volume of the ideal gas changes 10 times within the cycle? The working substance is nitrogen [$(10)^{0.4} = 2.5$].

(A) 40% (B) 25% (C) 60% (D) 75%

CTS199

ANSWER KEY

Question	1	2	3	4	5	6	7
Answer	C	B	D	ACD	A → (i, iii), B → (i), C → (ii)	B	B
Question	8	9	10	11			
Answer	2870	ABCD	B	C			

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